Crucible: Agentic Simulation Platform

Internal Proposal — March 2026

The Problem

Complex advisory engagements increasingly demand answers to questions that static models cannot answer:

- How will this market respond to a regulatory shock over the next 18 months?
- What are the second- and third-order effects of a competitor's entry?
- How does an escalating conflict affect our client's supply chain under different resolution scenarios?

Current approaches rely on expert judgment, spreadsheet models, or off-the-shelf tools that lack domain depth. They are slow to build, hard to update, and impossible to calibrate against live events.

We need a better way to model the world.

The Solution

Crucible is a platform for **agent-based simulation** — the gold standard methodology for modeling complex adaptive systems. It combines:

- **Domain-specific theory** (economics, geopolitics, organizational behavior, market dynamics) to structure each simulation
- **Real-time research ingestion** (academic literature, economic data feeds, news) to calibrate models against the world as it is
- **Agentic scaffolding** — an AI-driven intake process that designs a simulation from a consultant's plain-language description, with minimal manual configuration

The result: a consultant describes a scenario on Monday. By Tuesday, a calibrated, running simulation is delivering insights.

How It Works

1. Scenario Intake (The Forge)

A consultant opens the Crucible Forge and describes a scenario in plain language:

> *"Model the competitive dynamics in the US generic pharmaceuticals market following proposed FDA biosimilar approval reforms."*

A research-grounded scoping agent immediately:

- Searches academic literature (arXiv, SSRN) for relevant theory
- Pulls calibration data (FRED, World Bank, industry feeds)
- Asks targeted, informed follow-up questions based on what it finds
- Iterates with the consultant to refine scope, actors, and key uncertainties

The agent builds a **SimSpec** — a structured scenario definition — in the background. When complete, it hands off automatically.

2. Simulation Build (Automated)

The platform selects appropriate theoretical models from a curated library, configures agent-based actors with Bayesian belief systems, and runs the simulation. No manual parameter entry. No bespoke code per engagement.

3. Live Dashboard

Results are delivered through a clean, interactive war-room dashboard:

- Key performance indicators and scenario trajectories
- Named snapshots at critical decision points

- Narrative feed explaining what the simulation is doing and why
- Replay controls to explore alternative paths

4. Continuous Calibration

Once running, the simulation stays current. A background data feed agent monitors relevant news and economic indicators, adjusts parameters automatically, and flags significant developments to the consultant team.

Theory Library

Crucible ships with a curated library of validated formal models across five domains. Each theory is implemented as a composable module: it exposes a standard interface (parameters, state variables, update rules) that the Theory Mapper can wire into any simulation without bespoke code.

Domain 1: Geopolitics & Conflict

Model	Description	Key Parameters
Richardson Arms Race	Differential equation model of mutual military buildup driven by threat perception and grievance. Classic dyadic escalation baseline.	Fatigue coefficient, threat sensitivity, grievance load
Wittman-Zartman Negotiation	Models ripeness for conflict termination — identifies when both parties perceive a mutually hurting stalemate and a negotiated exit becomes rational.	Stalemate threshold, cost tolerance, perceived BATNA
Fearon Bargaining (DIA)	Explains why rational actors fight. Models private information asymmetries, commitment problems, and issue indivisibility as drivers of conflict onset and duration.	Information gap, credibility discount, issue divisibility
Prospect Theory (Kahneman-Tversky applied to IR)	Captures loss-aversion in state decision-making — actors in the domain of losses take greater risks than classical expected utility predicts.	Reference point, loss-aversion coefficient
Selectorate Theory (Bueno de Mesquita)	Models leader survival logic — how the size of the winning coalition shapes foreign policy decisions, war initiation, and concession behavior.	Winning coalition size, selectorate size, loyalty norm

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Domain 2: Market & Competitive Dynamics

Model	Description	Key Parameters
Porter's Five Forces	Structural analysis of industry attractiveness — maps rivalry, substitution, buyer/supplier power, and entry threat as continuous pressure variables.	Entry barrier height, switching costs, concentration ratio
Cournot / Bertrand Competition	Oligopoly pricing and quantity dynamics. Cournot models quantity competition; Bertrand models price competition to marginal cost.	Number of firms, marginal cost, demand elasticity
Supply/Demand Shock Propagation	Models how upstream disruptions (supply shocks) or demand collapses ripple through a market system with inventory buffers and price stickiness.	Shock magnitude, buffer depth, price rigidity
Financial Contagion (DebtRank)	Network-based model of systemic risk propagation across financially linked institutions or markets.	Exposure matrix, leverage ratios, threshold sensitivity

Bass Diffusion Model	Forecasts adoption curves for new products or technologies — separates innovator-driven early adoption from imitator-driven mass market growth.	Innovation coefficient (p), imitation coefficient (q)
Hotelling Spatial Competition	Models firm positioning along a product or policy dimension when consumers prefer proximity — explains clustering and differentiation decisions.	Transport cost, consumer distribution, entry cost

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Domain 3: Organizational & Corporate Behavior

Model	Description	Key Parameters
Principal-Agent Theory	Models misaligned incentives between principals (owners, regulators, boards) and agents (managers, employees). Drives hidden action and adverse selection dynamics.	Monitoring cost, agent risk aversion, incentive alignment
Institutional Theory (DiMaggio & Powell)	Explains organizational isomorphism — why firms in a sector converge on similar structures through coercive, mimetic, and normative pressures.	Regulatory pressure, peer visibility, legitimacy pressure
Diffusion of Innovation (Rogers)	Models how a practice, technology, or policy spreads through a population segmented into innovators, early adopters, majority, and laggards.	Adoption thresholds per segment, network density
Organizational Resilience (Weick HRO)	High-Reliability Organization model — captures how organizations maintain function under stress through redundancy, deference to expertise, and sense-making loops.	Redundancy factor, expertise centralization, failure rate
Stakeholder Salience (Mitchell et al.)	Prioritizes which actors receive organizational attention based on power, legitimacy, and urgency — useful for modeling multi-stakeholder scenarios.	Salience scores, urgency decay, power distribution

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Domain 4: Macroeconomics & Policy

Model	Description	Key Parameters
Keynesian Multiplier	Models the amplified effect of a fiscal stimulus or shock on aggregate demand through successive rounds of spending.	Marginal propensity to consume, leakage rates
Regulatory Shock Propagation	Captures how a policy change (new regulation, tariff, license requirement) transmits through a sector via compliance cost, market exit, and behavioral adjustment.	Compliance cost distribution, exit threshold, adjustment speed
IS-LM / AS-AD Framework	Standard macroeconomic equilibrium model linking output, interest rates, and price levels — useful for monetary and fiscal policy scenario analysis.	Investment sensitivity, money demand elasticity, price flexibility
Debt Sustainability (DSA)	Models sovereign or corporate debt trajectory under different growth, interest rate, and primary balance scenarios — standard IMF/World Bank methodology.	Debt/GDP ratio, interest-growth differential, primary balance
Input-Output (Leontief)	Sector-level model of economic interdependence — traces how a disruption in one sector propagates through supply chains across the whole economy.	Technical coefficients matrix, final demand vector

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Domain 5: Social & Behavioral Dynamics

Model	Description	Key Parameters
Schelling Segregation	Demonstrates how mild individual preferences produce strong macro-level segregation — applicable to market segmentation, political polarization, and urban dynamics.	Tolerance threshold, population mix
Opinion Dynamics (Deffuant-Weisbuch)	Models belief convergence and polarization in a population — agents update opinions only when the gap with interacting agents falls below a confidence threshold.	Confidence threshold, convergence rate, network topology
SIR / SEIR Contagion	Epidemiological spread model adapted for idea diffusion, crisis contagion, financial panic, or social movement propagation.	Transmission rate (β), recovery rate (γ), network structure
Tipping Point / Threshold Models (Granovetter)	Models cascade phenomena — strikes, protests, bank runs, market crashes — as sequences of agents crossing individual action thresholds.	Threshold distribution, network connectivity

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Composability

Models are not used in isolation. The Theory Mapper composes them into layered simulations:

- A **geopolitical scenario** might layer Richardson (escalation) + Fearon (onset logic) + Prospect Theory (decision bias) + Input-Output (economic damage propagation)
- A **market entry scenario** might layer Porter's Five Forces (structural baseline) + Bass Diffusion (adoption) + Hotelling (positioning) + Principal-Agent (partner dynamics)
- A **regulatory scenario** might layer Regulatory Shock Propagation + Keynesian Multiplier + Institutional Theory (industry response)

The library grows with each engagement. New theories are formalized and added as reusable modules after each project.

Use Cases

Crucible is sector-agnostic. Initial target domains:

- **Geopolitical risk** — conflict escalation, sanctions, trade disruption
- **Competitive intelligence** — market entry, price wars, M&A dynamics
- **Regulatory scenarios** — policy reform, compliance shocks, market restructuring
- **Supply chain resilience** — disruption modeling, sourcing alternatives
- **Organizational change** — merger integration, restructuring, leadership transition

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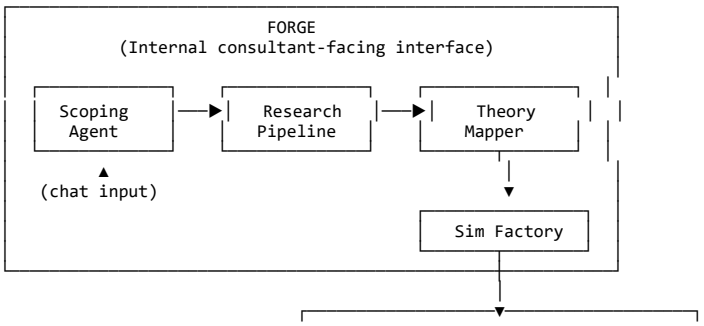
Platform Architecture

Crucible is a cloud-native platform with five distinct subsystems. Each is independently deployable and communicates through well-defined interfaces.

System Overview



```
┌──────────────────────────────────────────┐
│              SIMULATION ENGINE           │
│   BDI Agents + Theory Modules + Clock    │
└──────────────────────────────────────────┘
                    │
     ┌──────────────┼──────────────┐
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┌──────────────┐ ┌──────────────┐ ┌──────────────┐
│ FastAPI Backend │ │ Data Feed Agent │ │ Snapshot Store │
│ (REST + WS)  │ │ (live calibration) │ │ (named + scheduled) │
└──────────────┘ └──────────────┘ └──────────────┘
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│                FRONTEND                  │
│ ForgePage │ DashboardPage │ PortalPage   │
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```

Subsystem 1: Scoping Agent

The entry point. A conversational LLM agent (Claude API) that transforms a free-text scenario description into a structured **SimSpec** object.

Behavior:

- On first input, fires all research adapters in parallel (background, non-blocking)
- Asks informed questions grounded in what research returns — not a static intake form
- Iteratively refines scope: actors, timeframe, key uncertainties, outcome metrics
- Builds the SimSpec object conversationally, invisible to the user
- Hands off to Theory Mapper when SimSpec is complete

Key design principles:

- Research precedes questions — the agent never asks something it could have looked up
- Each conversation turn is context-aware: later questions incorporate earlier answers and new research results
- The consultant sees a natural dialogue; the system is assembling a structured object

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Subsystem 2: Research Pipeline

A set of lightweight adapters that fetch, parse, and summarize external data sources on demand. Used by the Scoping Agent at intake and by the Data Feed Agent at runtime.

Adapter	Source	Use
`arxiv_adapter`	arXiv API	Academic papers on relevant theory and domain
`ssrn_adapter`	SSRN search	Economics, finance, social science preprints
`fred_adapter`	FRED API	Macro economic time series (GDP, inflation, rates, trade)
`worldbank_adapter`	World Bank API	Development indicators, country-level data
`news_adapter`	RSS / OSINT feeds	Live event tracking, calibration triggers

Each adapter returns a normalized `ResearchResult` object (source, title, summary, relevance score, raw data). The Scoping Agent and Theory Mapper consume these uniformly.

Subsystem 3: SimSpec

The structured output of the Scoping Agent. SimSpec is the contract between intake and simulation — it fully describes a scenario in machine-readable form.

```
SimSpec
├── scenario_id          UUID
├── title                str
├── domain               enum: geopolitics | market | org | macro | social | composite
├── actors[]
│   ├── id               str
│   ├── type             enum: state | firm | regulator | population | coalition
│   ├── beliefs{}        dict: variable → initial probability distribution
│   ├── desires[]        list: goal objects with priority weights
│   └── capabilities{}   dict: action → capacity value
├── environment{}        dict: shared world-state variables
├── theories[]           list: theory module IDs selected by Theory Mapper
```

```
├── parameters{}         dict: theory param → value (from research + defaults)
├── timeframe
│   ├── start            date
│   ├── end              date
│   └── tick_unit        enum: day | week | month | quarter
├── uncertainties[]      list: variables with probability distributions
├── outcome_metrics[]    list: KPI definitions for the dashboard
└── research_sources[]   list: ResearchResult references used in construction
```

SimSpec is serialized to JSON and stored — it is the reproducible definition of a simulation. Any simulation can be reconstructed from its SimSpec.

Subsystem 4: Simulation Engine

The core runtime. Domain-agnostic. Executes the scenario defined in a SimSpec.

Agent Architecture (BDI):

Each actor in the SimSpec becomes a BDI (Belief-Desire-Intention) agent at runtime:

- **Beliefs** — probability distributions over world-state variables, updated each tick via Bayesian inference
- **Desires** — weighted goal set (e.g., maximize territorial control, minimize economic damage, maintain coalition support)
- **Intentions** — current action plan, selected by a decision function that maximizes expected utility given beliefs and desires
- **Actions** — discrete moves (escalate, negotiate, sanction, invest, exit, etc.) with effects on the shared environment

Tick Loop:

```
for each tick:
    1. Environment updated (external shocks, data feed inputs)
    2. Each agent observes environment → updates beliefs (Bayesian)
    3. Each agent evaluates intentions against updated beliefs
    4. Each agent selects and executes action
    5. Actions resolve → environment updated
    6. Theory modules apply update rules (e.g., Richardson equations, multiplier effects)
    7. Outcome metrics recorded
    8. Snapshot trigger evaluated (scheduled or threshold-based)
```

Theory Module Interface:

Every theory module exposes:

- `parameters: dict` — configurable inputs
- `state_variables: dict` — variables it reads from and writes to environment
- `update(env, agents, tick) → env` — pure function applied each tick

This makes modules composable without modification.

Subsystem 5: Data Feed Agent

A background agent that runs while a simulation is live. Monitors real-world signals and recalibrates simulation parameters automatically.

Behavior:

- Polls news and data adapters on a configurable schedule
- Passes new information to Claude API for relevance assessment and parameter delta estimation
- Applies approved parameter adjustments to the running simulation
- Logs each calibration event with source, reasoning, and delta applied
- Flags high-significance events to the consultant dashboard as alerts

This is the "stays current" capability — a simulation launched on Monday reflects Tuesday's events by Wednesday.

Subsystem 6: API & Data Layer

FastAPI backend exposes:

- `POST /simulations` — create simulation from SimSpec
- `GET /simulations/{id}/state` — current simulation state
- `GET /simulations/{id}/metrics` — outcome KPI time series
- `POST /simulations/{id}/snapshots` — name and save a snapshot
- `GET /simulations/{id}/snapshots` — list snapshots
- `WS /simulations/{id}/stream` — real-time tick stream for dashboard

- `POST /forge/intake` — submit scenario text to Scoping Agent
- `GET /forge/intake/{session_id}` — poll scoping session state

Data persistence:

- Simulation state: serialized per-tick to append-only log
- Snapshots: named JSON blobs (SimSpec + state at point-in-time)
- Research cache: TTL-keyed per adapter + query hash

Subsystem 7: Frontend

Three distinct views, unified codebase:

ForgePage — Scoping Agent UI

- Chat interface with streaming responses
- Background research status indicators (live as adapters return)
- SimSpec preview panel (builds in real-time as conversation progresses)
- "Launch simulation" CTA when SimSpec is complete

DashboardPage — Internal war-room view

- Live KPI panels with tick-by-tick updates (WebSocket)
- Agent belief and intention state viewer
- Timeline with named snapshot markers
- Replay controls — step forward/backward, branch from any snapshot
- Calibration event log (Data Feed Agent activity)
- Theory module parameter viewer

PortalPage — Client-facing SaaS view

- Clean, branded layout — no simulation machinery visible
- Key findings panels with narrative explanations
- Snapshot comparison (before/after scenarios)
- Export: PDF report, data CSV, shareable link

Infrastructure

Component	Technology	Deployment
Backend	Python 3.12 / FastAPI	Railway (Dockerfile)
Simulation engine	Pure Python (asyncio)	Co-located with backend
Frontend	React 19 / Vite / TypeScript	Vercel
Database	PostgreSQL (Railway managed)	Railway
Cache	Redis (research results, session state)	Railway
AI	Claude API (claude-sonnet-4-6)	Anthropic
Scheduling	APScheduler	In-process

This pattern is proven — it is exactly the architecture of the Hormuz proof of concept, operating in production.

Proof of Concept

An internal proof of concept — the **Hormuz Crisis Simulation** — has already validated the core architecture:

- 18 BDI (Belief-Desire-Intention) agents with Bayesian belief updates
- Richardson escalation dynamics and dual termination theory branches
- OSINT scraping pipeline feeding live parameter adjustments
- Full war-room dashboard with replay, named snapshots, and narrative feed
- Deployed and running on cloud infrastructure

This is not a concept. The engine works. Crucible is the platform that makes it deployable across engagements.

Build Plan

Phase 1 — Foundation (Weeks 1–3)

- Core simulation engine (generalized from Hormuz proof of concept)
- SimSpec schema and theory library (4 domains, 10+ models)
- Research adapters: arXiv, FRED, World Bank, news
- Hormuz ported as reference scenario #1

Phase 2 — Scoping Agent (Weeks 4–7)

- Conversational intake UI (Forge)
- Background research pipeline wired to adapters
- SimSpec builder with theory mapper
- End-to-end: plain-language description → running simulation

Phase 3 — Client Portal & Pilot (Weeks 8–12)

- Branded client-facing dashboard
- Snapshot management and export
- Continuous calibration agent (live data feeds)
- Deploy on a live engagement — target turnaround: 48 hours from brief to running sim
- Second reference scenario built from pilot engagement

Next Steps

1. Align leadership on platform vision
2. Identify pilot engagement candidate
3. Begin Phase 1 development
4. Reconvene in 4 weeks with working demo

Prepared March 2026